

Subject Description Form

Subject Code	ITC4202T
Subject Title	Smart Textiles for Wearable Applications
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Exclusion SFT403FI Smart Textiles for Wearable Applications
Objectives	The subject provides the knowledge of the latest development, including materials, technologies and products, in smart textiles for wearable application.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) assess technologies involved in smart textile and wearable products and manufacturing processes, their advantages and disadvantages; (b) describe the development trends in technologies for smart textiles and wearable products and their potential in industrial applications. (c) cooperate efficiently in a team to achieve goals; (d) develop critical and creative thinking; (e) develop the ability to adapt new technologies and update their knowledge; (f) develop the ability for sourcing, selecting and integrating information.
Subject Synopsis/ Indicative Syllabus	(I) Smart Fibres and Fibrous Assemblies Electric functions of textile polymers. Electrically conductive fibres, optical fibres. Photochromic fibres. Electrochromic fabrics. Active polymer fibres. (II) Interactive Textile Devices Fibre-based electronic circuit and systems. Flexible fabric strain sensors. Soft tactile sensors, textile electrodes for biopotentials. Textile energy harvesters. Textile energy storages. Photonic fabric displays. Textile circuit boards. (III) Wearable Products and System Analysis Thermal regulating garments. Health monitoring garments. shoes. Wearable phototherapy device. (IV) Manufacturing Technologies Smart yarn formations. Incorporation of smart functions by weaving, knitting, embroidery, printing, coating and lamination.

Teaching/Learning Methodology	Lectures will be used to introduce the principles, applications and decision implications of various technological developments to students. Laboratory/studio involving case studies will be used to provide hands-on experience and supplement the lecture program. Students will be encouraged to interact with the lecturer, and present their own findings based on investigation of the new developments and new products. Individual assignments, laboratory/studio work and group projects will be the major parts of continuous assessment components.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>1. <i>Assignment/ laboratory work</i></td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. <i>Group project</i></td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td>✓</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: In the continuous assessment, by individual assignments, laboratory/studio work and group projects, all rounded education will be focused. In addition to professional knowledge, the importance of teamwork, critical and creative thinking, the ability to adapt new technologies, and the demonstration of initiative in sourcing, selecting and integrating information will be stressed. For example, interaction among group members will improve their communication skills, their teamwork spirit, leadership as well as entrepreneurship potential. In order to gain systematic technical knowledge/skills, students will also be asked to do assignments.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	Continuous Assessment	50%	✓	✓	✓	✓	✓	✓	1. <i>Assignment/ laboratory work</i>	25%	✓	✓	✓	✓	✓	✓	2. <i>Group project</i>	25%	✓	✓	✓	✓	✓	✓	Examination	50%	✓	✓		✓			Total	100%						
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	In the examination, students' learning performance in terms of defining concepts, explaining principles and describing and analysing application examples as well as providing various forms of rationale for smart textile and wearable products and corresponding integration processes will be assessed.	
Student Study Effort Expected	Class contact:	
	• Lecture	26 Hrs.
	• Laboratory	12 Hrs.
	Other student study effort:	
	• Assignments	25 Hrs.
	• Self-study	42 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Books</u> <u>Essentials</u> Tao, X. M. (2015), <i>Handbook of smart textiles</i>. Springer. (e-version available) Tao, X. M. (2005), <i>Wearable electronics and photonics</i>. Elsevier. Schneegass, S., & Amft, O. (2017). <i>Smart textiles: Fundamentals, design, and interaction (Human-computer interaction series)</i>. Springer. <u>Supplementary</u> Tao, X. M. (2001), <i>Smart fibres, fabrics and clothing – fundamentals and applications</i>. Elsevier. Helal, A. (Sumi). (2008), Mokhtari, Mounir, Abdulrazak. <i>The Engineering Handbook of Smart Technology for Aging, disability, and Independence</i>, John Wiley & Sons, Inc., Bessam. (e-version available)	

	<u>Periodicals</u> Textile Research Journal Smart Materials Bulletin Smart Materials and Structures High Performance Textiles Textile Outlook International Medical Textiles
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